

Modeling predictions of Respiratory Failure using Data Collected from Clinical Trial on neuromuscular blockade reversal drugs in Kidney Transplantation surgery

Francesca Kitch, Mateo Candelaria, Kai Weiner, Carolina Arroyo, Daniela Luque-Sanchez and Georgia Reed (Mentor: Alex Kaizer)

Prediction Models Background

- Predictive models are built using combinations of covariates to predict an outcome.
- The goal of our prediction models is to maximize Area Under the Curve (AUC) which is a combination of maximizing sensitivity (detection of outcome/true positive rate) and specificity (correct detection of the outcome/true negatives).
- Machine Learning methodologies can be used to build classification models that can predict respiratory failure based on explanatory variables
- The performance of these models can be evaluated using metrics of AUC, accuracy, sensitivity, and specificity

Logistic Regression

- Logistic Regression allows for the prediction modeling of binary outcomes (i.e. respiratory failure)
- Singular predictors and interaction term models were explored

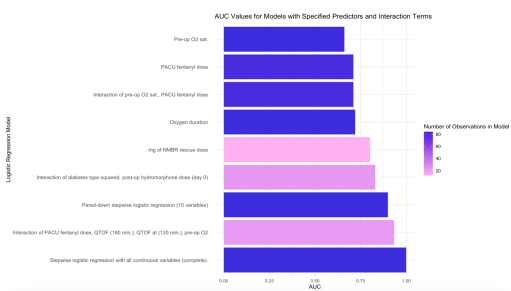
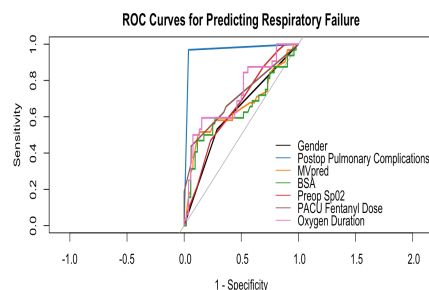


Figure 1. AUC and number of observations for most successful models



Stepwise Logistic Regression

- The stepAIC function automates variable selection, running through all possible models and selects one that minimizes AIC
- Model 1: Model selected using all non-collinear, continuous variables
- Model 2: Model selected by looking at the trace of the step function, choosing a model with 10 covariates
- Additional models were created with binary predictors but reported insignificant predictors

Table 1. Stepwise Models and reported values

qTOF Backslide

- Predict backslide in recovery from NMBR (declining recovery from anesthesia)
- qTOF measured ~ every 30 minutes for 3 hours
 - Measure of nerve response to electrode stimulation
- Identified 30-minute time periods in minute-by-minute ventilator data corresponding to qTOF measurements present in REDCap patient information
- Utilized split linear regression to parameterize subsets of ventilator data
 - Error in estimated tidal volume, respiratory rate, observed adjusted tidal volume
- Applied logistic regression to split regression parameters in an attempt to predict backslide
 - Parameters include: beta coefficient with largest magnitude, range of explanatory variable, and ideal number of splits.

K Nearest Neighbors

- K Nearest Neighbors (KNN) is a supervised machine learning algorithm that classifies observations based on their similarity to their k nearest neighbors in the training set data.
- The 84 observations were randomly split into training, validation, and testing sets with 50, 10, and 24 samples, respectively.
- An exhaustive search for the best K was conducted using the validation set. This search resulted in a validation accuracy of 100% when K =8.
- Using K =8, the test set was introduced to the model and yielded an accuracy of 91.67%, with a sensitivity of 77.78% and a specificity of 100%

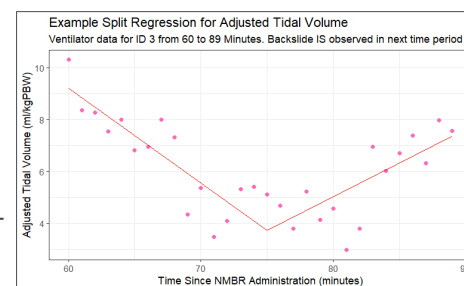


Figure 2. Graph of qTOF Split Regression for Adjusted Tidal Volume

Limitations and Next Steps

- The small sample size of this data lead to risk of overfitting of the models
- Next steps would include simulating the models to calculate their bias

Acknowledgements

Dr. Ana Fernandez-Bustamante formulated the dataset and design of the study, and Dr. Alex Kaizer mentored the statistical modeling and analysis process

Modeling predictions of Respiratory Failure using Data Collected from Clinical Trial: Sugammadex vs Neostigmine for Kidney Transplantation

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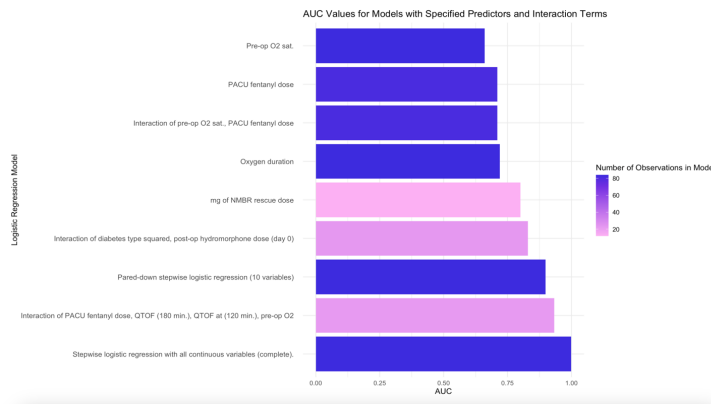


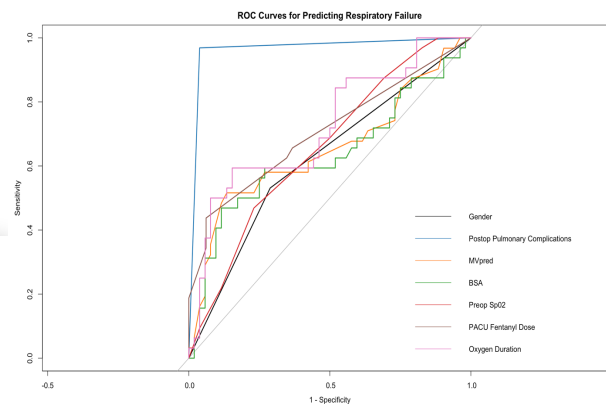
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Model	AUC value	Number of Covariates
Model 1	1.0	57 (45 significant)
Model 2	0.8984	10 (6 significant)

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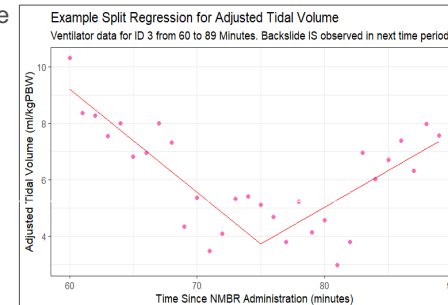


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Visualizations of Minute-by-Minute Data Collected From Clinical Trial: Sugammadex versus Neostigmine for Kidney Transplantation

Skylar Hsieh and Caroline Schwengler (Mentor: Alex Kaizer)

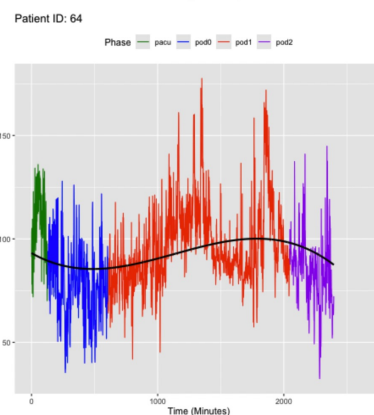
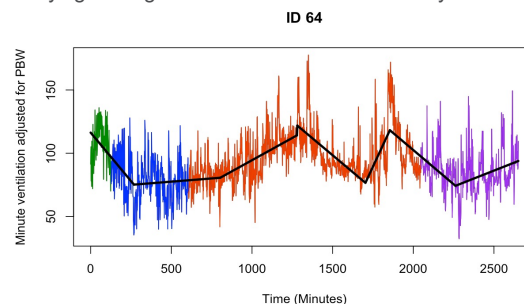
Background Information

The Kidney Transplantation Trial was a double-blind, randomized study that compared the efficiency of sugammadex versus neostigmine for reversal of neuromuscular blockade (NMBR) at the end of kidney transplantation in patients with severe kidney dysfunction. The original study focused on 84 patients and found that sugammadex did not necessarily provide improved NMBR compared to neostigmine in patients undergoing kidney transplantation.

Our goal, following this work, was to visualize the minute-by-minute ventilation data to observe trends that could potentially be of use in predicting kidney failure after transplantation.

Segmented Regression

- Difficult to follow or interpret minute-by-minute data
- Use to analyze trends too complicated for simple linear regression
- Two algorithms to determine the number of segments
- Cannot perform regression for patients with missing data (began imputation work)
- Overlaying the regressions does not illuminate any obvious trends

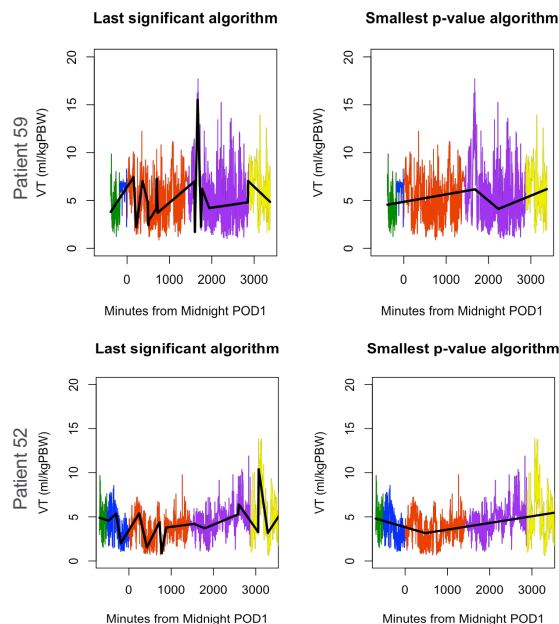
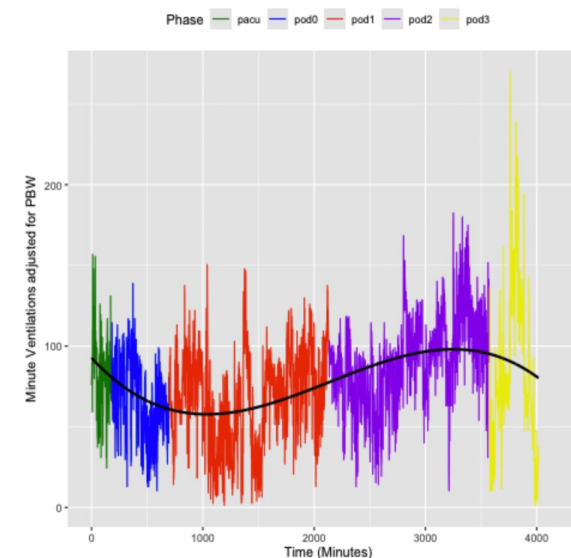


Animated Graphs with Splines

Animated plots for each patient ID with spline regression lines overlaid

- The plots show the minutes by the adjusted minute ventilation data by PBW (predicted body weight), colored by phase
- Spline lines are used to look at trends in non-linear data by using knots to fit a smooth regression line
- There are gaps between graphs for some of the patients because of missing ventilation data

Patient ID: 52



Both the segmented regression and spline lines show more interpretable trends of the data over time. For this patient we can see that they are predicting the same trends of initial decline before rising again. However the segmented regression pulls attention to a dip in the ventilation data that the spline line does feature. The main difference is in their future applications, while segmented regressions are more interpretable because we know the equation and can interpret the meaning of the coefficients. Splines do not require us to assume that our data is linear and are smoother which can make them more visually appealing.

https://docs.google.com/document/d/1Ct1-o8C5bQez6pNyavOwelaDcwUAT9sj7_MuKID7SZI/edit?usp=sharing